

Network analysis of stress and well-being among college students: A longitudinal study

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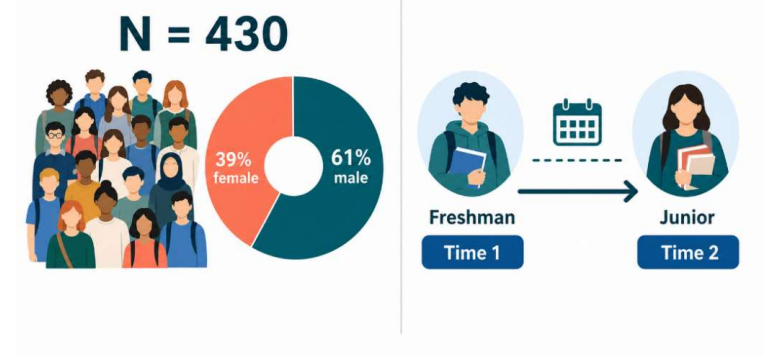
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Objective

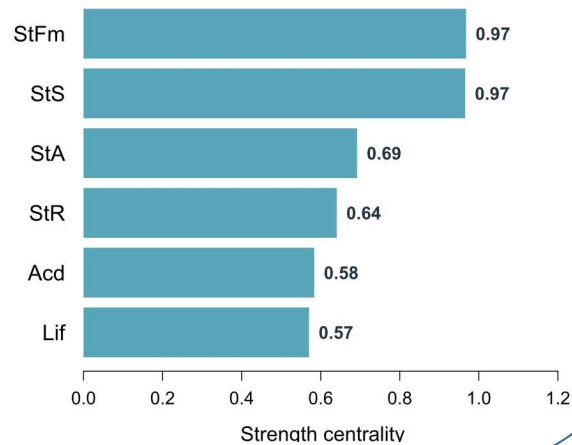
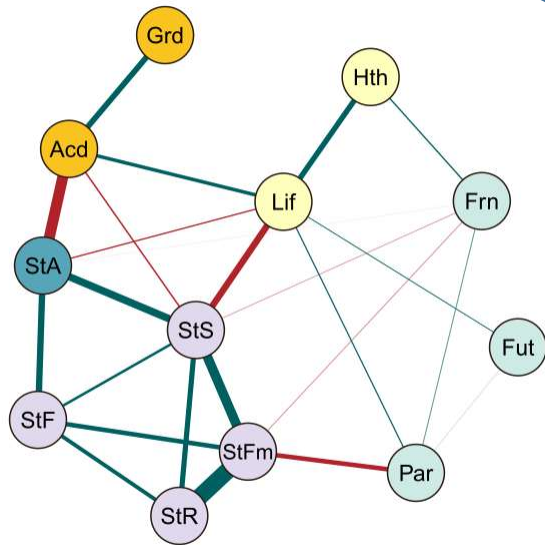
- College students experience interconnected academic, financial, family, and relationship-related stressors.
- We investigated
 - how stress and well-being were connected in freshman and junior years using network analysis
 - how these relationships changed from freshman to junior year

Methods

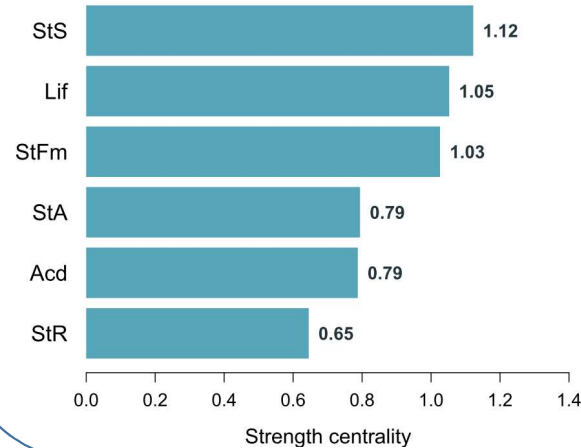
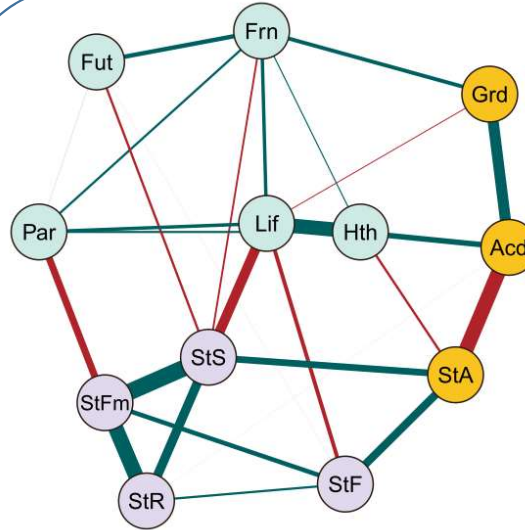
Network analysis examined connections among stressors (academic, financial, family, intimate-relationship, peer-related stress), academic performance, relationships, perceived health, future outlook, and life satisfaction.



Results Freshman Year



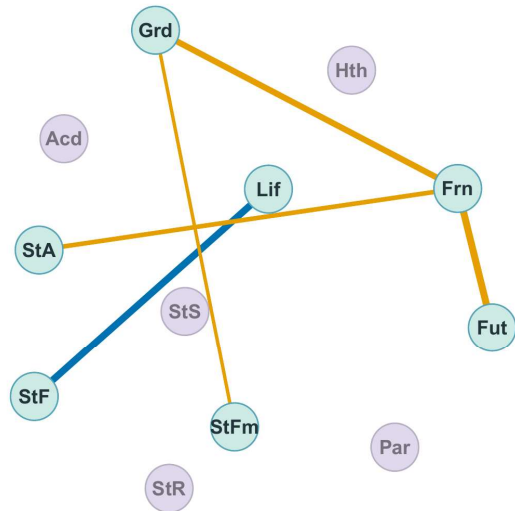
Junior Year



- Social stress was the most central node among the variables in both freshman and junior years.
- Family-relationship stress was also highly central during the freshman year.

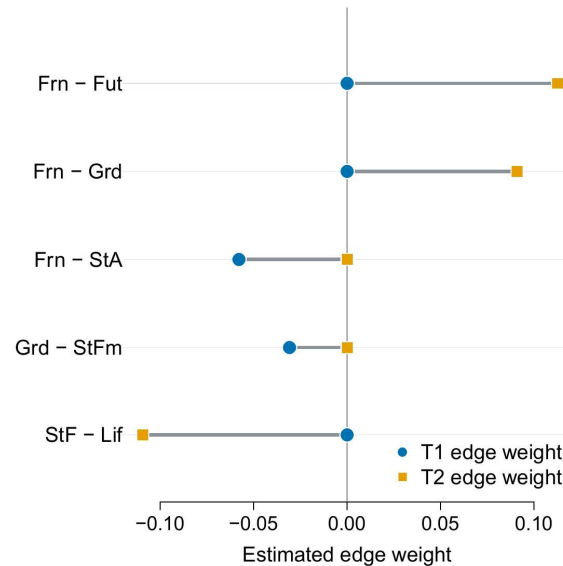
Comparison of junior and freshman years

A. Network location of significant differences



Only NCT edge-invariance tests with unadjusted $p < .05$ are shown; $\Delta = T2 - T1$.

B. T1 and T2 edge weights



- Overall network connectivity and structure remained stable.
- Exploratory edge-level results suggested close friendships had stronger links with grades and future outlook at T2.
- Negative Frn–StA and Grd–StFm associations were stronger at T1
- Financial stress became more negatively associated with life satisfaction at T2.

Discussion

- Although the overall network remained stable, the roles of close friendships and financial stress may shift as students progress through college.
- These findings suggest the value of sustained peer-support programs and increased financial guidance as students advance through college.
- Edge-level findings are exploratory and based on unadjusted p-values.

Guide to Network Nodes

Stress-related nodes

Higher scores indicate greater perceived stress.



- **StA** — Academic-performance stress



- **StF** — Financial stress



- **StFm** — Parent–adolescent relationship stress



- **StR** — Intimate-relationship stress



- **StS** — Peer-relationship stress

Academic, relationship, and well-being nodes



- **Acd** — Concern about academic performance
1 = least bothered; 5 = most bothered



- **Grd** — School grades
1 = low; 5 = high



- **Par** — Closeness with parents
Higher scores = closer relationship



- **Frn** — Having close friends
1 = no; 2 = not really; 3 = yes



- **Hth** — Perceived health
1 = poor; 5 = good



- **Fut** — Future planning
1 = low; 4 = high



- **Lif** — Feelings about life (life satisfaction)
Higher scores = happier and less distressed